

Assignment 1

PCCST501 – Computer Networks

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How the Internet Delivers a Web Page

A Case Study on Wikipedia (www.wikipedia.org)

1. Draw the TCP/IP protocol stack and explain the role of each layer.

The TCP/IP protocol suite is the foundation of communication on the Internet. It is a hierarchical set of protocols organized into five layers, where each layer is responsible for a specific part of the communication process and relies on the services provided by the layer below it.

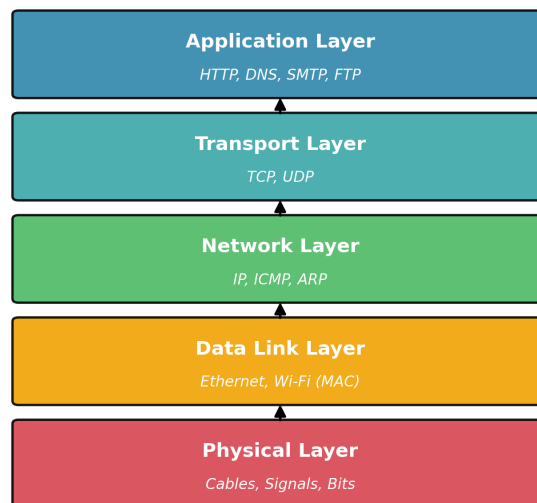


Figure 1: The five-layer TCP/IP protocol stack used for retrieving a web page.

Application Layer: Provides services directly to the user. When the user requests the Wikipedia homepage, protocols such as **HTTP** (to fetch the page) and **DNS** (to resolve the domain name) operate at this layer. It is responsible for process-to-process communication between the browser and the web server.

Transport Layer: Responsible for end-to-end delivery of data between the source and destination hosts. **TCP** is used here to provide reliable, connection-oriented, ordered delivery of the HTTP request and response, including flow control and error control.

Network Layer: Responsible for host-to-host delivery of packets (datagrams) across the internet using **IP addresses**. The **IP** protocol determines the best path/route for each packet from the browser's computer to the Wikipedia server, possibly through multiple routers.

Data Link Layer: Responsible for hop-to-hop delivery of frames across a single link (e.g., a LAN or Wi-Fi link) using physical (MAC) addresses. It takes the datagram from the network layer and encapsulates it into a frame for transmission across a specific link.

Physical Layer: Responsible for transmitting raw bits as electrical, optical, or radio signals across the physical transmission medium (cable, fibre, or air) connecting two adjacent devices.

2. Explain the role of DNS in converting the domain name into an IP address.

Computers communicate using numeric IP addresses, but humans find domain names such as **www.wikipedia.org** easier to remember. The **Domain Name System (DNS)** is a distributed, hierarchical directory service that maps human-readable domain names to their corresponding IP addresses.

When the user types **www.wikipedia.org** and presses Enter, the browser invokes a **DNS resolver (client)** on the local machine. If the mapping is not already cached, the resolver queries a local DNS server, which in turn may contact a **root server**, a **top-level domain (.org) server**, and finally the **authoritative name server** for **wikipedia.org** — either recursively or iteratively — until the IP address is found. This IP address is then returned to the browser, which uses it to establish a connection with the actual Wikipedia web server. The result is typically cached (using a Time-To-Live value) to speed up future requests to the same domain.

3. Describe how HTTP is used to request and receive the webpage.

Once the browser has the IP address of the Wikipedia server, it uses the **HyperText Transfer Protocol (HTTP)** to communicate with the server. HTTP operates on a request-response model over a TCP connection, typically using well-known port 80 (or port 443 for HTTPS).

The browser (HTTP client) sends a **GET** request message containing the request line (method, URL, and HTTP version) along with header lines specifying details such as the accepted content types and the browser type. The Wikipedia server (HTTP server) processes this request and returns a **response message** consisting of a status line (e.g., "200 OK"), response headers (such as content type and length), and the body containing the actual web page content (HTML, CSS, JavaScript, images, etc.). Because a typical Wikipedia article page contains many embedded objects (images, stylesheets, scripts), the browser issues multiple such GET requests, generally reusing a single **persistent HTTP/1.1 connection** to retrieve them efficiently.

4. Explain how TCP provides reliable communication for HTTP.

HTTP relies on the **Transmission Control Protocol (TCP)** at the transport layer because TCP is **connection-oriented** and **reliable**. Before any HTTP message is exchanged, the browser and the Wikipedia server perform a **three-way TCP handshake** (SYN, SYN-ACK,

ACK) to establish a logical connection and agree on parameters such as buffer size and initial sequence numbers.

TCP ensures reliability for HTTP communication through the following mechanisms:

- **Error control:** TCP uses sequence numbers and acknowledgements to detect lost or corrupted segments and retransmits them if needed, ensuring the complete web page data arrives intact.
- **Ordered delivery:** Since a web page may be split across multiple TCP segments, sequence numbers allow the receiving side to reassemble the segments in the correct order.
- **Flow control:** TCP matches the sender's data rate with the receiver's ability to process data, preventing the Wikipedia server from overwhelming the browser's buffer.
- **Congestion control:** TCP adjusts its sending rate based on network conditions to avoid congestion collapse on the path between client and server.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

Wikipedia follows the **Client–Server architecture**, not a Peer-to-Peer (P2P) architecture.

Justification:

- The **service provider** is a set of powerful, always-available Wikipedia/Wikimedia Foundation web servers (the server process) that continuously listen for and respond to requests.
- The **user's browser** acts purely as a client — it initiates a request only when needed and terminates once the required content (article, image, page) is received.
- All content (articles, images, revision history) is centrally hosted and served from the Wikimedia Foundation's data centres / Content Delivery Network (CDN) servers, rather than being shared directly between end-user devices as happens in P2P systems such as BitTorrent.
- The roles of client and server are fixed and cannot be interchanged — a browser cannot act as a Wikipedia server, which is a defining characteristic of the client-server paradigm.

6. Compare HTTP and FTP with at least four differences.

Although both HTTP and FTP are standard, client-server application-layer protocols built on top of TCP, they differ in purpose and operation as summarised below:

Basis	HTTP	FTP
Purpose	Used to request and transfer web pages and embedded web objects (HTML, images, scripts).	Used specifically for transferring (uploading/downloading) files between a client and server.

Connections	Uses a single TCP connection (persistent by default in HTTP/1.1) for both control and data.	Uses two separate TCP connections — a control connection (port 21) and a data connection (port 20).
State	Originally stateless; state is maintained artificially using cookies.	Maintains state throughout the session via the control connection (e.g., current directory).
Port Number	Well-known port 80 (HTTP) or 443 (HTTPS).	Well-known ports 21 (control) and 20 (data).
Data Handling	Optimised for small, quick request/response exchanges of mixed media.	Optimised for transferring large files efficiently, with support for ASCII, binary, and different transmission modes.

7. Explain how cookies improve user experience with one real-world example.

HTTP is inherently a **stateless** protocol — each request and response is treated independently, with no memory of prior interactions. **Cookies** were introduced to overcome this limitation by allowing a server to store small pieces of information about a client, which the browser sends back with every subsequent request to that server.

How it works: When the server first responds to a client, it includes a *Set-Cookie* header containing a unique identifier. The browser stores this cookie and automatically includes it in the *Cookie* header of all future requests to the same domain, allowing the server to recognise the returning user.

Real-world example (Wikipedia): When a user logs into Wikipedia and sets a preferred interface language or edits an article, a cookie stores information such as the session/login state and display preferences. On the next visit, this cookie allows Wikipedia to keep the user logged in automatically, remember the preferred language and reading list, and restore settings such as dark mode or font size — without requiring the user to re-enter this information every time. This significantly improves user experience by making browsing personalised and seamless.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back.

The figure below illustrates the complete communication flow — from the moment the user enters the URL in the browser, through DNS resolution and TCP connection establishment, to the HTTP request/response exchange and final rendering of the web page.

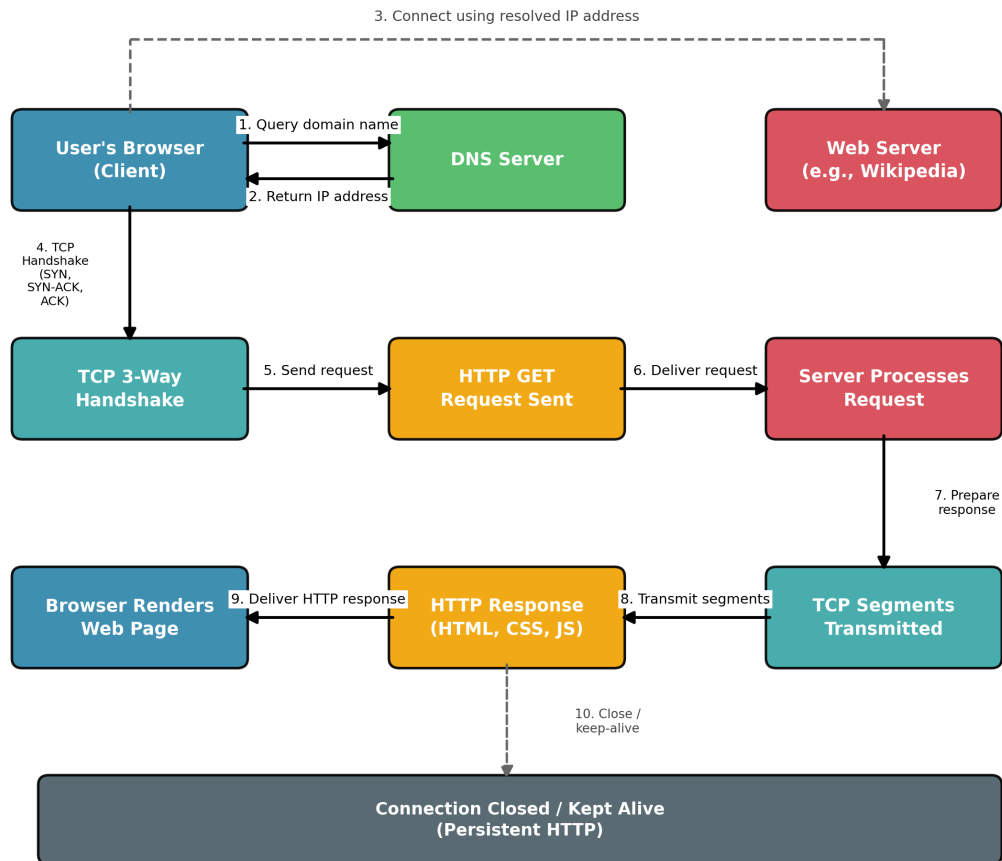


Figure 2: End-to-end communication flow between the browser (client) and the Wikipedia web server.

9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols form the backbone of everyday Internet usage, quietly working together to deliver a seamless browsing experience. As demonstrated through the example of accessing Wikipedia, a single user action — typing a URL — triggers a coordinated sequence involving DNS for name resolution, TCP for reliable transport, and HTTP for requesting and delivering content, all built upon the layered TCP/IP architecture. Without these protocols working in harmony, tasks such as browsing websites, reading articles, sending emails, or transferring files would be impossible. Furthermore, supporting mechanisms like cookies enhance usability by enabling personalisation and session continuity. Understanding this layered, protocol-driven process highlights how the Internet abstracts immense technical complexity into simple, reliable, and near-instantaneous services that billions of users depend on daily.